

Leggett & Platt Automotive

Functional Design Document

SCM FDD0603 – Sales Order Customs Broker Tracking

Prepared for

Leggett & Platt Automotive

Project

E03 - ERP Analysis, Design & Implementation

Work Stream

SCM

Prepared by:

Anupam Soni

Contributors:

Lacey Burnette, Andrew Jinks

Revision History and Signoff Sheet

Change Record:

Date	Author	Version	Change Reference
2019-03-19	Anupam Soni	1.0	First Draft
2019-03-22	Anupam Soni	1.1	Updated the following sections: 1. Assumptions and Business Rules 2. Business logic section for Broker ID on Packing Slip 3. Configuration section
2019-03-28	Anupam Soni	2.0	Assumption section has been updated

Client Review and Approval:

Name	Version Approved	Position	Date
Luke Bunch	1.0	LPA SCM Workstream Lead	2019-03-21
Luke Bunch	1.1	LPA SCM Workstream Lead	2019-03-22

Related Documents:

Document Name	Version	Date

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1 INTRODUCTION

1.1 PURPOSE

LPA needs the ability to assign a customs broker to every customer record to make sure it can track shipment related information at a customer level all the way down to a sales orders and resultant loads

The purpose of this document is to describe the functional implementation of the requirements identified by Leggett & Platt Automotive and Microsoft.

The Functional Design Document will cover the following design topics:

Design Topics	Description
Section 5.1	Modifications to Forms to support the functional design
Section 5.2	Modifications to Tables to support the functional design
Section 5.3	Modifications to Business Logic to support the functional design
Section 5.4	Modifications to List Pages to support the functional design
Section 6	Configurations and Security Considerations
Section 7	Test Cases to validate the modifications to support the functional design
Section 8	Appendix for supporting information and templates related to the functional design

1.2 ABBREVIATIONS AND GLOSSARY OF TERMS

Abbreviation	Explanation
Forms	The Forms used to enter data to support the process you are designing for.
Fast Tab	A FastTab is a container for data entry controls. A FastTab organizes fields into related chunks. FastTabs are used primarily on task pages such as details forms, dialog forms, simple list and details forms.
Tables	The tables that source the forms
Action Pane	An action pane is the part of a form that organizes and displays buttons that represent the actions the form supports.
List Pages	The list page presents the primary data of the application on a user interface that is optimized for browsing records, finding the right one, and then taking an action upon that record. The list page lets the user search, filter, sort, and preview the data.
Navigation Path	The standard navigation window presents the user with a module navigation control. This will indicate where the forms, list pages or periodic jobs are stored and how to navigate within D365.
String	Text Field

Data Source	Used for String fields that will point to data reference table, for example the field Delivery Terms is sourced from the table "DeliveryTerms".
Boolean	Yes or No value
Real	Used to indicate a Number or Currency field.

1.3 AUDIENCE / STAKEHOLDERS

Name	Position
Hollie Elliott	LPA Materials Manager
Stephanie Mitchell	LPA Plant Scheduler
Steve Shaften	LPA Logistics Supervisor
Mara Celic	LPA Purchasing Manager
Rolando Aviles	LPA Business Lead
Enrique Aldrete	LPA Solution Architect
Luke Bunch	LPA SCM Workstream Lead
Andrew Jinks	LPA SCM Business Analyst
Joseph Calicott	LPA SCM Business Analyst
Francis Moigula	Microsoft Solution Architect
Claudio Sbardella	Microsoft Delivery Architect
Ausif Hussain	Microsoft Technical Lead
Anupam Soni	Microsoft SCM Senior Consultant
Lacey Burnette	Microsoft SCM Consultant
Bob Fesmire	KPMG Business Architect
Shea Carroll	KPMG SCM Process Lead

1.4 FDD / DESIGN CLASSIFICATION

Design Classification	Included in Design? (Yes or No)
Creation or Modification to Dynamics Processes or Logic	Y
Customizations to Dynamics Forms / Screens	Y
Integration	N
Creating or Modifying Dynamics Reports or Inquiries	N

2 DESIGN SCOPE

2.1 GAP REQUIREMENTS

List the relevant gap requirements to be discussed in the scope of this FDD.

ID	Title
4462	Ability to select a customs broker for each customer

3 DESIGN ASSUMPTIONS & BUSINESS RULES

3.1 DESIGN ASSUMPTIONS

The following design assumptions have been made as part of this FDD

S. No.	Title
1.	The Broker ID field is a look-up field on the shipping carrier table in Standard D365. It is assumed that Custom Brokers would be set up in the same table as shipping carriers. This assumption has been verified by LPA SME's and the BA.
2.	The Shipping Carriers table will be used for maintaining both the carriers and customs brokers, and this applies to both the sales and purchasing side.
3.	In case of ISV (ATOS), the ISV solution will have to use the additional logic to ensure customs broker field is utilized on relevant EDI transactions. This is outside the scope of this design document, but it needs to be done during the ISV Design/Build phase. An issue has been logged in VSTS regarding the same.
4.	When a Sales Order is "released to warehouse" and an outbound shipping wave is created, the system requires the user to specify the Broker ID field manually on the outbound load. This is standard D365 functionality, and no change has been proposed to Transportation Management (in case we use loads and shipments in D365).
5.	Customs Brokers and Shipping Carriers will be maintained at each branch level (i.e. at a legal entity level in D365).
6.	The Broker ID field is not mandatory on the Customer Master, Sales Agreement, or Sales Order forms in D365.

3.2 BUSINESS RULES

The following business rules apply to this FDD

S. No.	Title
1.	The Broker ID field should copy over from the Customer form to the Sales Agreement form when a Sales Agreement is created for a Customer. It should then further copy over from the Sales Agreement form to the resultant Sales Order when a release sales order gets created in D365. If no sales agreements exist or are not created for a customer, but sales orders are being created directly, the system should still copy the customs broker field from the customer to the sales order.
2.	In D365, the Shipping Carrier table/form is in the Inventory Management Module, so from a data setup perspective the same needs to be configured in the Inventory

	Management Module. Please refer to the configuration section below in this document.
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4 BUSINESS PROCESS

4.1 PROCESS DEFINITION / BUSINESS PROCESS FLOW

This section is not applicable for this FDD.

4.1.1 Business Process Narrative

This section is not applicable for this FDD.

4.1.2 Business Process Flow

This section is not applicable for this FDD.

5 DESIGN

5.1 MODIFICATION TO FORMS

5.1.1 Customer Master Form

It is proposed to add the Broker ID field (which is present in Standard D365) to the Customer Master form. This field is present on the “All Sales Orders” form in Sales and Marketing Module as well as on the “All loads” form in Transportation Management Module.

ALL CUSTOMERS

0369-000001 : Magna - Ingram Windsor Seating

UPS zone

Tax exempt number

Delivery terms

Invoice attachment

No

FREIGHT

Freight accrual

No

Transportation

BILL OF LADING

Accepts express bill of lading

No

Broker ID

Shipping carrier

Name

BNSF Berlington North

CPU Customer Pick-U

Existing or New?	Existing
Form Name	CustTable
Form Label	All Customers
Navigation Path	Accounts Receivable > Customers > All Customers

Field Name	Control Name	Fast Tab	Field Type	Notes
Broker ID	TMCustTable_BrokerCode	Transportation	Look-up	Broker ID field, which is currently on the SO Header form. (TMSSalesTable_BrokerCode)

5.1.2 Sales Agreement Form

The Broker ID field should also be added on to the Sales Agreement form as shown below:

SALES AGREEMENTS

0369-000101 : 0369-000002 - Ford Motor Company - Ship to

Lines **Header** | On hold

0369-000002 | USD | 3/11/2019

General

CUSTOMER

Name: Ford Motor Company - Ship to

Customer account: 0369-000002

Contact:

Customer reference:

Customer requisition:

Invoice account: 0369-000002

Language: en-US

CURRENCY

Currency: USD

Fixed exchange rate:

DEFAULT VALIDITY PERIOD

Validity period for: Requested ship date

Effective date: 3/11/2019

Expiration date:

STATUS

Status: On hold

DOCUMENT

Sales agreement ID: 0369-000101

Document title:

Preparer: Anupam Soni

External document reference:

Sales agreement classification: Production Agreement

Default commitment: Product quantity commitment

Intercompany origin:

PROJECT

Project ID:

Broker ID:

Shipping carrier: BNSF, CPU, DHL, DHLG, MAERSK, OD

Name: Berlington North, Customer Pick-U, DHL - AIR, DHL - Ground, MAERSK, Old Dominion

Existing or New?	Existing
Form Name	Sales Agreements Header Form
Form Label	Sales Agreements
Navigation Path	Sales and Marketing > Sales Agreements > Sales Agreements

Field Name	Control Name	Fast Tab	Field Type	Notes
Broker ID	TMSSalesAgreement_BrokerCode	General	Look-up	Broker ID field, which is currently on the SO Header form. (TMSSalesTable_BrokerCode)

5.1.3 Packing Slip Journal Form

The Broker ID field should also be added on to the Packing Slip Journal form as shown below. This field will be shown on Posted Packing Slip Journal and will be un-editable on this form. The data on this field will be shown when the sales order packing slip is posted.

0369-000302 : MAGNA - INTEGRAM WINDSOR SEATING

Packing slip journal

Overview Lines Order tracking

CARRIER

Shipping carrier Carrier service Broker ID

TRACKING INFORMATION

Tracking number	Tracking URL
 We didn't find anything to show here.	

Existing or New?	Existing
Form Name	Packing Slip Journal Form
Form Label	Packing Slip Journal
Navigation Path	Sales and Marketing > Sales Orders > All Sales Orders

Field Name	Control Name	Fast Tab	Field Type	Notes
Broker ID	TMSSalesTable_BrokerCode	Order Tracking	Look-up	Once Packing Slip is posted for a SO, Broker ID field should show on the Posted Packing Slip Journal.

5.2 MODIFICATION TO TABLES

5.2.1 Customer Table

Existing or New?	Existing
Table Name	CustTable

Shared Table	No
--------------	----

Field Name	Field Type	Size	Notes
Broker ID (TMSCustTable_BrokerCode)	Look-up		Broker ID field, which is currently on the SO Header form. (TMSSalesTable_BrokerCode)

5.2.2 Sales Agreements

Existing or New?	Existing
Table Name	Sales Agreements
Form Label	Sales Agreements

Field Name	Field Type	Size	Notes
Broker ID (TMSSalesAgreement_BrokerCode)	Look-up		Broker ID field which is currently on the SO Header form. (TMSSalesTable_BrokerCode)

5.2.3 Packing Slip

Existing or New?	Existing
Table Name	CustPackingSlipJour
Form Label	Packing Slip Journal

Field Name	Field Type	Size	Notes
Broker ID (TMSSalesAgreement_BrokerCode)	Look-up		Broker ID field, which is currently on the SO Header form. (TMSSalesTable_BrokerCode)

5.3 MODIFICATION TO BUSINESS LOGIC

In case a Sales Order is being created using a Release Order from a Sales Agreement, the customs broker field will need to flow down from the Sales Agreement form to the Sales Order form as shown below:

The image displays two screenshots from the Dynamics 365 Finance and Operations interface, illustrating the flow of the customs broker field from a Sales Agreement to a Sales Order.

Top Screenshot: Sales Agreement Form

- Navigation:** Sales and marketing > Sales agreements > Sales agreements
- Form Header:** SALES AGREEMENTS, 0369-000071 : 0369-000001 - Magna - Ingram Windsor Seating
- General Tab:** Displays customer information (Magna - Ingram Windsor Sea...), currency (CAD), effective date (9/19/2018), and status (Effective).
- Delivery Address:** Magna - Ingram Windsor Sea... 201 Patillo Rd. R.R.#1 Tecumseh, ON N8N 2L9
- Customs Broker Field:** A dropdown menu is shown with the following options: BNSF, CPU, DHL, DHLG, MAERSK, OD, Berlington North, Customer Pick-Up, DHL - AIR, DHL - Ground, MAERSK, and Old Dominion.

Bottom Screenshot: Sales Order Form

- Navigation:** Sales and marketing > Sales orders > All sales orders
- Form Header:** SALES ORDER DETAILS, 0369-000532 : Magna - Ingram Windsor Seating
- Delivery Tab:** Displays delivery information, including transport (UPS zone), carrier information (Residential destination: No), and delivery date (Requested ship date: 3/21/2019).
- Transportation Tab:** Displays transportation information, including shipping carrier (BNSF), carrier service (Customer Pick-Up), and carrier group (DHL - AIR).
- Customs Broker Field:** A dropdown menu is shown with the following options: BNSF, CPU, DHL, DHLG, MAERSK, OD, Berlington Northern - Sante Fe, Customer Pick-Up, DHL - AIR, DHL - Ground, MAERSK, and Old Dominion.

A blue arrow points from the customs broker field in the Sales Agreement form to the customs broker field in the Sales Order form, indicating the flow of data.

The customs broker field should populate from the Customer Master table/form to the sales order header form as soon as a sales order is created for a customer in D365. This is applicable for cases in which a sales order is created directly for a customer and no valid sales agreement exist for customer in the system:

ALL CUSTOMERS

0369-000001 : Magna - Ingram Windsor Seating

UPS zone

Delivery terms

Tax exempt number

elinvoice attachment

Freight accrual

Freight

No

Transportation

BILL OF LADING

Accepts express bill of lading

No

Broker ID

BNSF

CPU

Berlington North

Customer Pick-U

Dynamics 365

Finance and Operations

Sales and marketing > Sales orders > All sales orders

0369

Save

New

Delete

SALES ORDER

SELL

MANAGE

PICK AND PACK

INVOICE

RETAIL

GENERAL

WAREHOUSE

TRANSPORTATION

OPTIONS

LOADS

Load planning workbench

ACTIONS

Release to warehouse

SALES ORDER DETAILS

0369-000532 : Magna - Ingram Windsor Seating

Lines

Header

Open order

Delivery

TRANSPORT

UPS zone

Call tag type

DELIVERY DATE

Requested ship date

Requested receipt date

Simulate delivery dates

Confirmed ship date

Confirmed receipt date

Delivery date control

Blind shipment

CARRIER INFORMATION

Residential destination

Ship complete

Automatic notification and cancellati...

Expedite

MISC. DELIVERY INFO

Mode of delivery

Delivery terms

Delivery reason

Reason for export

TRANSPORTATION

Shipping carrier

Carrier service

Carrier group

Mode

Route plan

Transportation template ID

Price and discount

Intercompany settings

Foreign trade

Broker ID

BNSF

CPU

DHL

DHLG

MAERSK

OD

Berlington Northern - Sante Fe

Customer Pick-Up

DHL - AIR

DHL - Ground

MAERSK

Old Dominion

The Broker ID field should also be added to the Posted Packing Slip Journal form. When the sales order packing slip is posted, the information from the sales order form should go to the order tracking section of the posted Packing slip journal form as shown below:

0369-000302 : MAGNA - INTEGRAM WINDSOR SEATING

Packing slip journal

Overview

Lines

Order tracking

CARRIER

Shipping carrier

Carrier service

Broker ID

TRACKING INFORMATION

Tracking number

Tracking URL



We didn't find anything to show here.

Existing or New?	Existing
Table Name	CustPackingSlipJour
Form Label	Packing Slip Journal

Field Name	Control Name	Fast Tab	Field Type	Notes
Broker ID	TMSSalesTable_BrokerCode	Order Tracking	Look-up	Once Packing Slip is posted for a SO, Broker ID field should show on the Posted Packing Slip Journal.

5.4 MODIFICATION TO LIST PAGES

It should be possible to add the Broker ID field on the list pages of Sales Agreements and Sales Orders. This is required to ensure that data can be filtered in different ways by using Broker ID field.

The list pages would then subsequently be used to create Tiles and should be able to be placed on different workspaces in D365 by individual users.

5.4.1 Sales Agreement List Page

The Broker ID field should be added to the Sales Agreements List Page. It should be possible to use this field to filter and sort data in this form. It should also be possible to use this field for the purpose of creating a Tile and place it on any appropriate workspace in D365 by the user:

SALES AGREEMENTS							
Filter							
✓ Sales agreem...	↑ Customer account	Name	Status	Broker ID	Document title	Currency	Agreement clas
0369-000001	0369-000002	Ford Motor Company - Ship to	Effective			USD	Production A
0369-000011	0369-000004	IMA - Detroit	Effective			USD	Production A
0369-000012	0369-000002	Ford Motor Company - Ship to	Effective			USD	Production A
0369-000023	0369-000002	Ford Motor Company - Ship to	Effective			USD	Production A
0369-000031	0369-000002	Ford Motor Company - Ship to	On hold			USD	Service Agre
0369-000051	0369-000003	Magna - Mississauga Stg. Syste...	On hold			USD	Production A
0369-000061	C00000003	Magna Seating Detroit	Effective			USD	Production A
0369-000071	0369-000001	Magna - Integram Windsor Sea...	Effective			CAD	Production A
0369-000081	0369-000001	Magna - Integram Windsor Sea...	On hold			CAD	Production A
0369-000091	0369-000003	Magna - Mississauga Stg. Syste...	On hold			USD	Production A
0369-000092	0369-000003	Magna - Mississauga Stg. Syste...	Effective			USD	Production A
0369-000101	0369-000002	Ford Motor Company - Ship to	On hold			USD	Production A

Existing or New?	Existing
List Page Name	Sales Agreements
Navigation Path	Sales and Marketing > Sales Agreements > Sales Agreements

Field Name	Control Name	Field Type	Notes
SalesAgreementListPage	TMSSalesAgreement_BrokerCode	Look-up	Broker ID field, which is currently on the SO Header form. (TMSSalesTable_BrokerCode)

5.4.2 Sales Orders List Page

The Broker ID field should be added to the Sales Orders List Page. It should be possible to use this field to filter and sort data in this form. It should also be possible to use this field for the purpose of creating a Tile and place it on any appropriate workspace in D365 by the user:

ALL SALES ORDERS									
Filter									
✓ Sales order ↑	Customer account	Customer name	Order type	Invoice account	Status	Release status	Broker ID	H...	Sales taker ▾
0369-000391	0369-000001	Magna - Ingram Windsor Sea...	Journal	0369-000001	Open order	Open			Anupam Soni
0369-000403	0369-000001	Magna - Ingram Windsor Sea...	Sales order	0369-000001	Open order	Open			Anupam Soni
0369-000422	0369-000002	Ford Motor Company - Ship to	Sales order	0369-000002	Invoiced	Released			Anupam Soni
0369-000423	0369-000002	Ford Motor Company - Ship to	Sales order	0369-000002	Invoiced	Released			Anupam Soni
0369-000431	0369-000001	Magna - Ingram Windsor Sea...	Sales order	0369-000001	Open order	Open			Anupam Soni
0369-000452	0369-000001	Magna - Ingram Windsor Sea...	Returned o...	0369-000001	Open order	Open			Anupam Soni
0369-000462	0369-000001	Magna - Ingram Windsor Sea...	Sales order	0369-000001	Open order	Open			Anupam Soni
0369-000511	0369-000003	Magna - Mississauga Stg. Syste...	Sales order	0369-000003	Open order	Open			Anupam Soni

Existing or New?	Existing
List Page Name	All Sales Orders
Navigation Path	Sales and Marketing > Sales Orders > All Sales Orders

Field Name	Control Name	Field Type	Notes
SalesTableListPage	TMSSalesTable_BrokerCode	Look-up	Broker ID field, which is currently on the SO Header form. (TMSSalesTable_BrokerCode)

6 CONFIGURATIONS / PARAMETERS / SECURITY CONSIDERATIONS

6.1 SHIPPING CARRIERS FORM

The screenshot below shows the shipping carrier form and table. This is the same table where we will be setting up Brokers as well. Just as in the example shown below, the 'Mode' field will be utilized to specify if the record in the table is a Broker or not:

The screenshot displays the 'SHIPPING CARRIERS' form in SAP S/4HANA. The form is titled 'SHIPPING CARRIERS' and includes a left sidebar with a filter and a list of carriers. The main form area has tabs for 'Overview', 'Carrier fuel indexes', 'Carrier accessorial charges', 'Accessorial assignments', 'Constraints', 'Miscellaneous charge', and 'OPTIONS'. The 'Overview' tab is active, showing fields for 'Activate shipping carrier' (Yes/No), 'Tracking URL', 'Transportation tender type' (None), 'Pro number sequence', 'Website URL', 'Vendor account', 'SCAC', and 'Activate carrier rating' (No). Below the form is a table titled 'Services' with columns: Carrier service, Name, Load template ID, Transportation method, Mode of delivery, and External code. The table has one row with 'Ocean' as the carrier service, 'OCEAN' as the transportation method, and 'Canad-Ocea' as the mode of delivery.

Existing or New?	Existing
Form Name	TMSCarrier
Form Label	Shipping Carriers
Navigation Path	Inventory Management > Set up > Shipping Carriers > Shipping Carriers

6.1.1 Modifications to Parameter Forms

This section is not applicable for this FDD.

6.1.2 Modifications to Parameter Table

This section is not applicable for this FDD.

6.2 SECURITY ROLE CONSIDERATIONS

This section is not applicable for this FDD.

7 ERROR HANDLING

This section is not applicable for this FDD.

Error #	Error Description	Infolog Error Message

8 TEST CASES

Given below are the Test Cases that are applicable for this FDD.

8.1 TEST CASES

VSTS ID	Test Case Title
13860	Create a stand-alone sales order and enter a customs broker on the sales order
13863	Create SA with customs broker selected then create an SO off the SA then change the customs broker on the SA and ensure that the existing SO doesn't change and that any new SOs created have the updated customs broker value
13861	Edit the customs broker value on a sales agreement and a sales order and ensure it can be edited or changed
13862	Ensure the customs broker field can be queried against and tiles created using a filter on this field
13859	Enter a customs broker on an SA and then create an SO off of the SA to ensure the customs broker defaults onto the SO

9 APPENDICES

This section is not applicable for this FDD.